

1. Administrative & Study Identification

Full Study Title: A Clinical Study to Evaluate the Effect of Renal Impairment on the Single-Dose Pharmacokinetics of MK-1084
Short Title/Acronym: A Clinical Study of MK-1084 in Participants With Renal Impairment (MK-1084-010) **Protocol Number:** MK-1084-010
NCT Number: NCT06814119 **Sponsor Name:** Merck Sharp & Dohme LLC
Investigational Product: MK-1084 (Calderasib) **Phase of Development:** Phase 1 **Medical Monitor:** Clinical Operations Lead, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate what happens to MK-1084 levels in the body over time (single-dose pharmacokinetics) when given to participants with moderate or severe renal impairment (RI) as compared to participants who are in good health. **Secondary Objectives:** To evaluate the safety and tolerability of a single dose of MK-1084 in people with moderate or severe RI and healthy control subjects.

2.2 Background & Rationale MK-1084 (calderasib) is an investigational KRAS G12C oral inhibitor being evaluated for the treatment of advanced solid tumors. Because cancer patients may present with co-morbid renal dysfunction, it is necessary to determine how compromised kidney function affects the absorption, distribution, metabolism, and excretion of MK-1084. This study aims to provide pharmacokinetic and safety data to guide safe and accurate dosing of MK-1084 in patients with moderate to severe renal impairment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Control (Healthy participants with normal renal function) **Blinding:** Open-label **Randomization:** Non-randomized (cohort assignment based on renal function status)

3.2 Planned Enrollment & Duration Target \$N\$: 34 evaluable participants **Number of Sites:** Multiple clinical research centers in the United States (e.g., Orlando, FL; Chicago, IL) **Estimated Study Start:** Early 2025 **Estimated Primary Completion:** Mid-to-Late 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants between 18 and 75 years of age. 2. Body Mass Index (BMI) ≥ 18.0 and ≤ 40.0 kg/m². 3. **Participants with severe or moderate RI:** With the exception of RI, is sufficiently healthy for study participation; has stable renal function with no clinically significant change in renal status at least 29 days prior to dosing, and is not currently or has not been previously on dialysis for at least 1 year. 4. **Participants with normal renal function:** Is medically healthy.

4.2 Exclusion Criteria 1. Has a history of cancer (malignancy). 2. Has positive results for human immunodeficiency virus (HIV), hepatitis B surface antigen, or hepatitis C virus. 3. Has had major surgery and/or donated or lost a significant volume of blood within 56 days prior to dosing. 4. **Participants with severe or moderate RI:** Has a history or presence of renal artery stenosis; has a renal transplant or nephrectomy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of MK-1084 **Route of Administration:** Oral **Duration of Treatment:** Single administration with post-dose observation and follow-up sampling.

5.2 Concomitant & Prohibited Therapy Permitted: Standard treatments for underlying conditions contributing to stable renal impairment, provided they do not interact with study drug metabolism. **Prohibited:** Concomitant use of strong or moderate inhibitors/inducers of CYP enzymes that may confound the pharmacokinetic measurements of MK-1084.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to Day -1	Informed Consent, Medical History, Labs, Renal Function Assessment
Baseline / Dosing	Day 1	Confinement, Single Dose Administration of MK-1084, Baseline PK/Safety
Observation	Post-Dose	Serial blood collections for PK parameters, AE assessments, Vital Signs
End of Study	Follow-Up	Final PK/Safety Labs, General Physical Examination, Exit procedures

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetic profile of MK-1084 following a single oral dose, specifically measuring Area Under the Concentration-time Curve (AUC) and Maximum

Concentration (Cmax) over time. **Measurement Method:** Serial blood sampling (plasma concentration) and central laboratory bioanalytical quantification.

7.2 Secondary & Exploratory Endpoints * Safety and tolerability evaluated by the incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine assessment of AEs through clinical laboratory tests, physical examinations, and vital signs at specified intervals. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review mechanisms managed by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set for primary endpoints; All-Subjects-as-Treated (Safety Set) for AE evaluation. **Sample Size Justification:** \$N=34\$ participants properly powered to characterize PK differences across varying degrees of renal impairment vs. normal renal function. **Handling of Missing Data:** Standard non-compartmental PK analysis guidelines will be employed (e.g., omitting missing intervals from AUC calculations).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite and remote monitoring visits to assure adherence to PK sampling times and accurate documentation of adverse events. **Audit Trail:** Validated eCRF tracking with secure data clarification processes.

11. Study Identifiers & Governance

Primary ID: MK-1084-010 **Registry Number:** NCT06814119 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of MK-1084 in Participants With Renal Impairment (MK-1084-010) **Official Regulatory Title:** A Clinical Study to Evaluate the Effect of Renal Impairment on the Single-Dose Pharmacokinetics of MK-1084

12. Clinical Framework (Renal Impairment Specific)

Primary Condition: Renal Impairment (Moderate and Severe)
Diagnosis Threshold: Stable renal function assessed by eGFR classifications, with no clinically significant changes for at least 29 days. **Medical Methodology:** Pharmacokinetic profiling.
Optimization Target: Identifying the impact of renal impairment on MK-1084 clearance to guide future clinical dosing regimens.

13. Study Procedures & Methodology

Management Pathway: Controlled single-dose inpatient administration with intensive plasma sampling. **Education Protocol (HCP):** Site personnel training on precise timing requirements for PK sampling and sample handling procedures. **Education Protocol (Patient):** Informed consent detailing required clinical confinement, blood draw schedule, and potential side effects. **Quality Audit Mechanism:** Source data verification emphasizing PK blood collection timestamps and comprehensive safety reporting.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6 to 9 weeks (screening through final follow-up). **Onsite Visit Cadence:** Confinement period for intensive initial dosing and serial sampling, followed by specific outpatient PK visits. **Remote Monitoring Frequency:** Standard check-ins as clinically indicated post-confinement. **Checklist Review Interval:** Regular review of concomitant medications and adverse events throughout the study's duration.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				